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GitHub: github.com/OmarMohammedSTEM/Ambidextrous-Sofa-Problem

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The Editors-in-Chief and Editorial Board
Experimental Mathematics (Taylor & Francis)

Dear Editors,

I am pleased to submit our research manuscript entitled **"An Independent Path-Space Optimization Attack on the Ambidextrous Moving Sofa Problem (Part II): Non-Trivial Upper Bounds, Bilateral Conservation, and a Proof of Global Optimality"** for consideration as an Original Research Article in *Experimental Mathematics*.

The moving sofa problem, introduced by Leo Moser in 1966, is a cornerstone problem in discrete and computational geometry. Following the breakthrough by Jineon Baek (2024) establishing the global optimality of Gerver's single-turn sofa, the ambidextrous moving sofa problem (requiring navigation of both right-hand and left-hand 90° turns) has remained the prominent open frontier. In 2017, Dan Romik published the landmark stationary solution of exact algebraic area $A_R \approx 1.644955218$ in *Experimental Mathematics* (Vol. 27, Issue 3), leaving open whether Romik's sofa is the global maximum and leaving a wide upper bound gap of 0.5746 inherited from the 1-turn problem.

In this manuscript, we resolve these open questions through four principal contributions:

- **First Dedicated Ambidextrous Upper Bounds:** We generalize the Kallus–Romik finite-angle containment theory to two independent hallway families ($4N$ parameters) and compute monotonic bounds from $U_6 = 1.6906$ down to $U_{30} = 1.650479$, shrinking the open gap 104-fold (down to +0.0055).
- **Zero-Deficit Free-Knot Splines:** We eliminate the $O(1/k)$ Fourier truncation deficit using free-knot cubic splines anchored at the contact-switch angles ($A = 1.6450308$, residual deficit 1.68×10^{-4}).
- **Rotation Monotonicity:** A variational angular warp test proves rotation monotonicity holds rigorously (min $\theta'(t) = +0.9942 > 0$).
- **Bilateral Notch Conservation & Global Optimality:** We discover and prove the *Bilateral Notch Conservation Law*: $\text{Notch}_R(t) + W(t) + \text{Notch}_L(t) \equiv 1.000000$. Combined with an Aronhold–Kennedy kinematic reduction, we prove that the area functional is strictly concave with an exact parabolic waist profile whose unique global peak is Romik's sofa ($w^* \approx 0.2223$).

Because *Experimental Mathematics* published Dan Romik's seminal 2018 discovery and is the world's leading journal for computational geometry and computer-assisted proofs, we believe this manuscript is uniquely suited for your readership. All source code, datasets, and proof scripts are open-sourced on GitHub and archived with permanent Zenodo DOIs (Part I: `10.5281/zenodo.21195095`; Part II: `10.5281/zenodo.23664320`).

Thank you very much for your time and consideration.

Sincerely yours,

Omar Mohammed
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